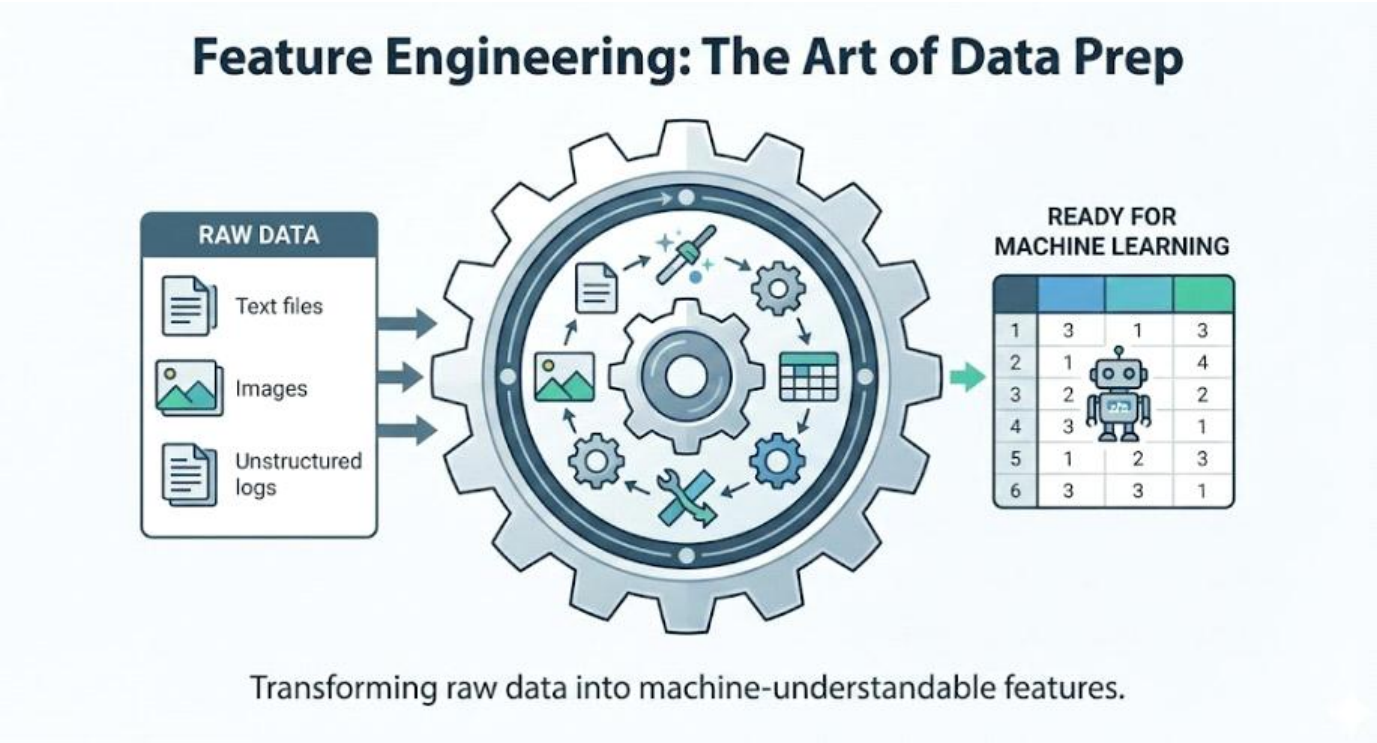


Encoding Categorical Data

08 February 2026 10:56



Feature Transformation: A Key Step

BEFORE: Mixed Data Types			
ID	Age	Color	Value
1	23	Red	450
2	24	Blue	970
3	25	Green	170

AFTER: Transformed Features			
ID	Age_Scaled	Color_Encoded	Value_Log
1	0.625	1	0.4097
2	0.679	2	0.6042
3	0.804	3	1.3967

Understanding Categorical Data

CATEGORICAL DATA
(Strings)

Nominal
(No Order)

Red, Blue, Green



Red, Blue, Green

Ordinal
(Has Order)

Small, Medium, Large



Small, Medium, Large

MACHINE-READABLE
(Numbers)

Color_ID	Size_ID
1	0
2	1
3	1
3	2
0	1



One-Hot Encoding: A Technique for Nominal Data

BEFORE: Single Categorical Column

Color
Red
Blue
Green



AFTER: Multiple Binary Columns

Is_Red	Is_Blue	Is_Green
1	0	0
0	1	0
0	1	0

Creates new binary columns for each category.



Label encoding

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INPUT DATA (X)

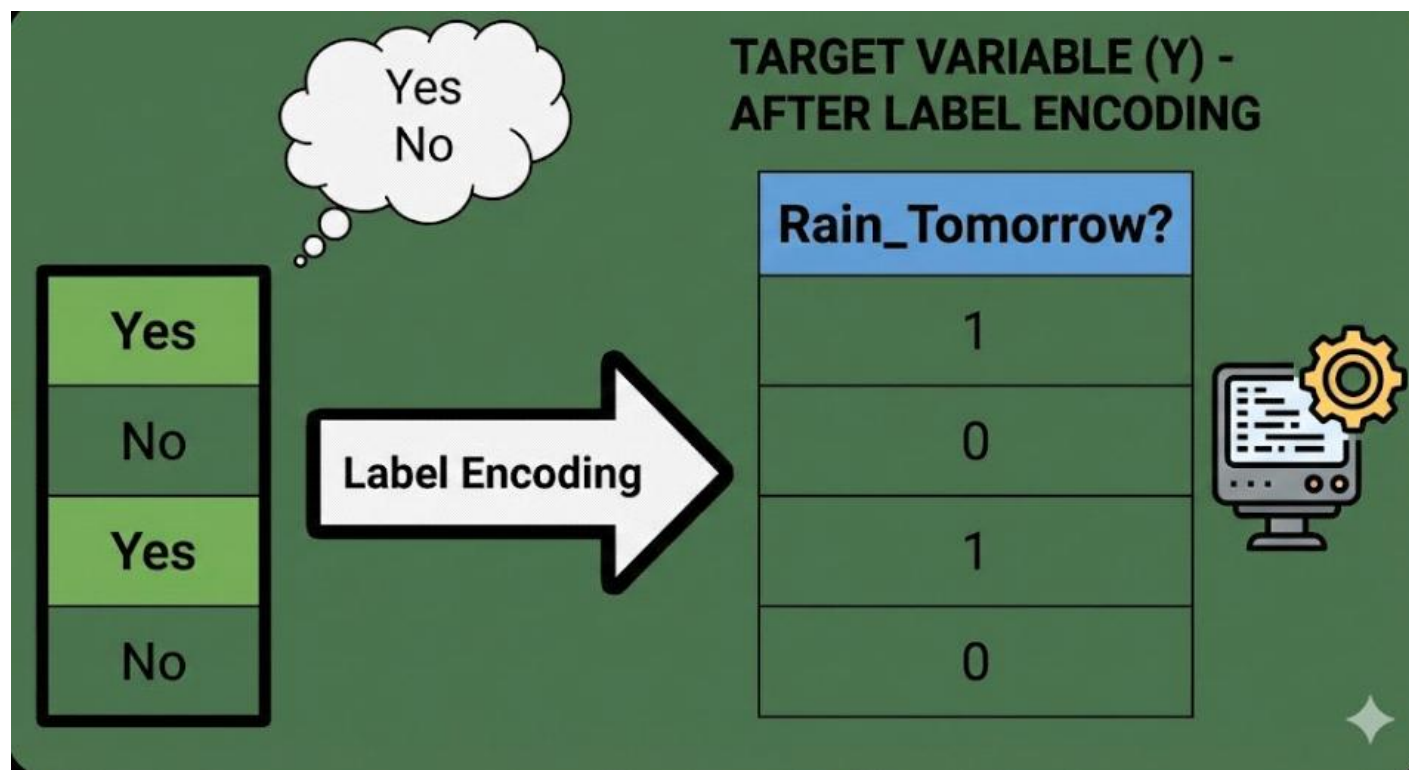
Temperature (C)	Humidity (%)
25	70
30	60
22	80
24	90

...

TARGET VARIABLE (Y)

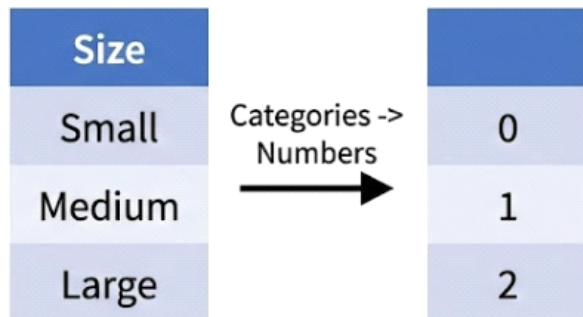
Rain_Tomorrow?
Yes
No
Yes
No

...

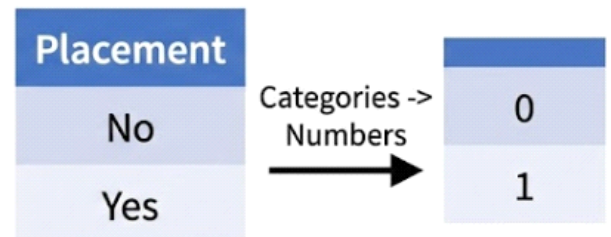


Ordinal Encoding vs. Label Encoding

ORDINAL ENCODING (For Input Features X)



LABEL ENCODING (For Target Variable Y)



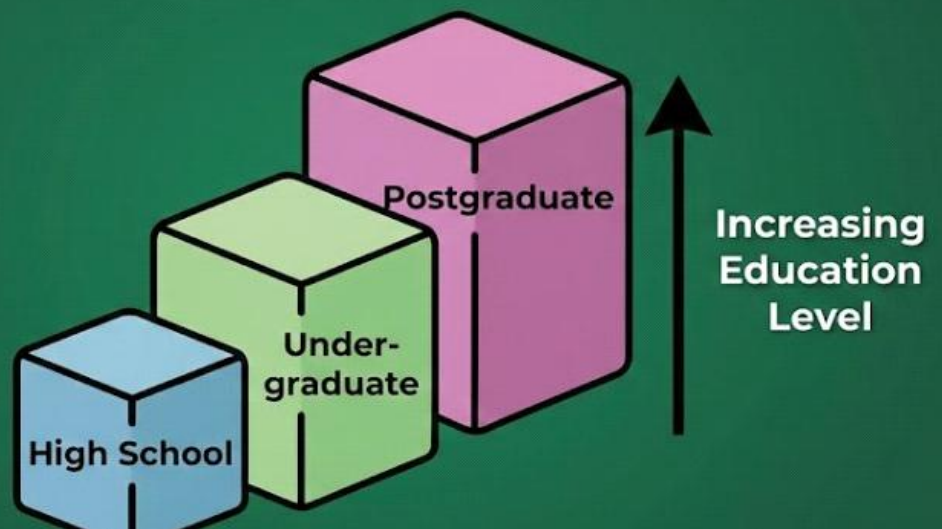
How Ordinal encoding works.

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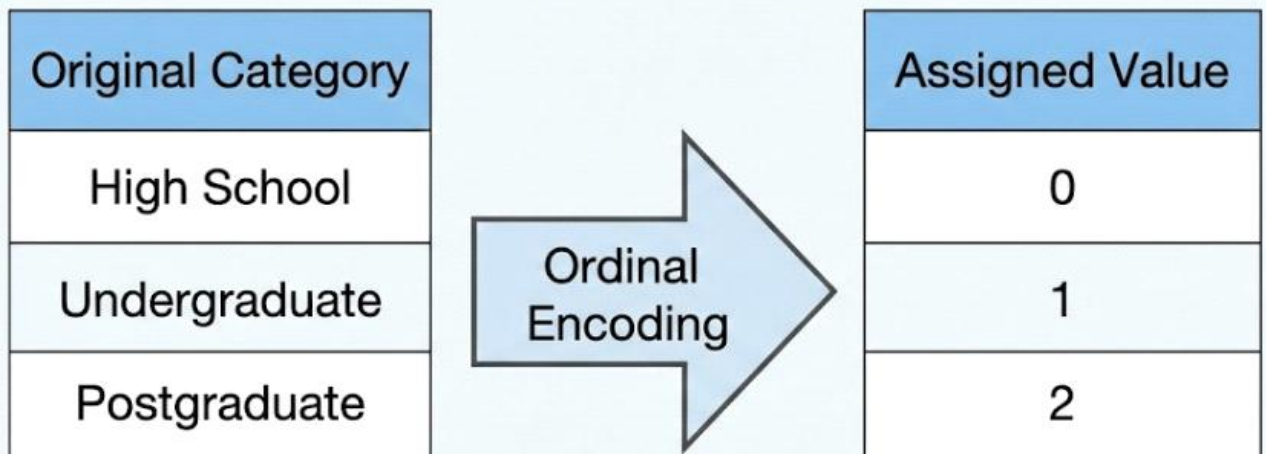
Original Categorical Data (Strings)

Education
High School
Undergraduate
Postgraduate
Undergraduate
High School
Postgraduate

Understanding Ordinal Data: There is an Order!



The Transformation: Assigning Numbers

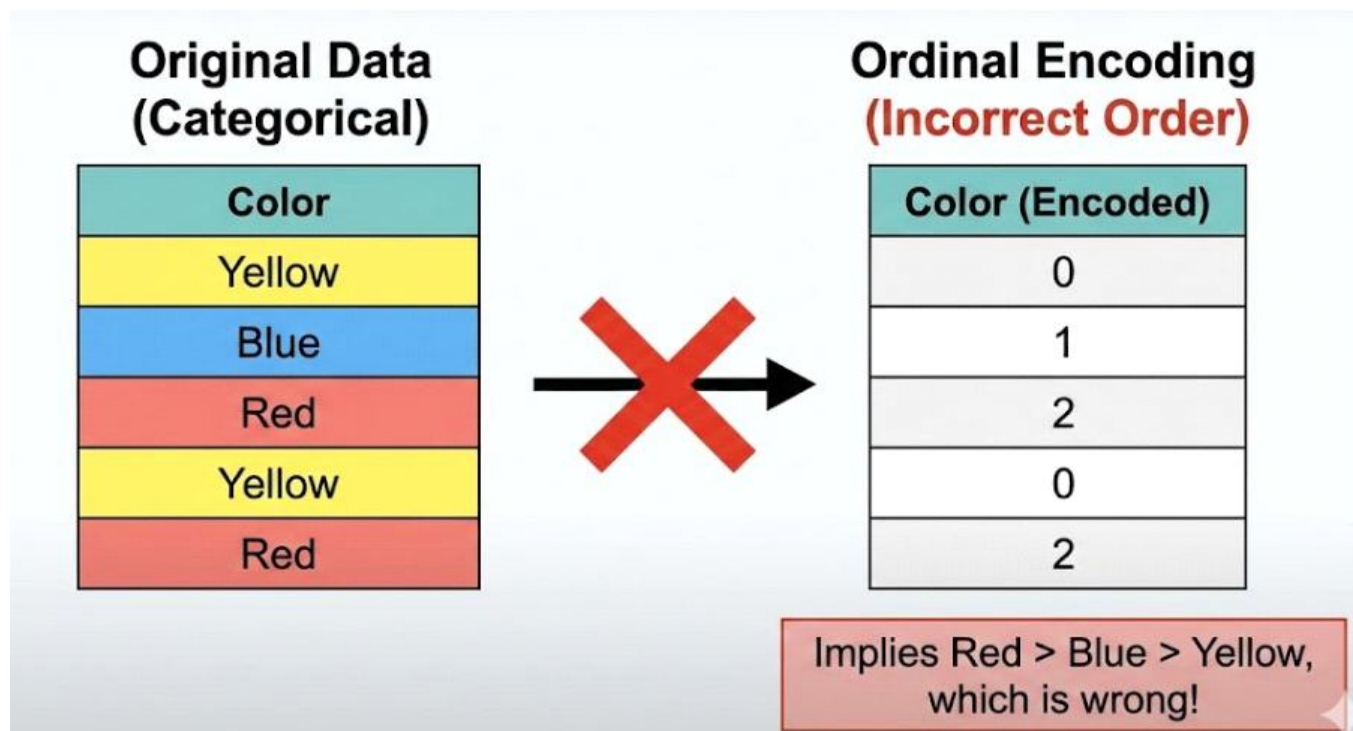


Transformed Data (Numbers)

Education	Education (Encoded) ✓
High School	0
Undergraduate	1
Postgraduate	2
Undergraduate	1
High School	0
Postgraduate	2

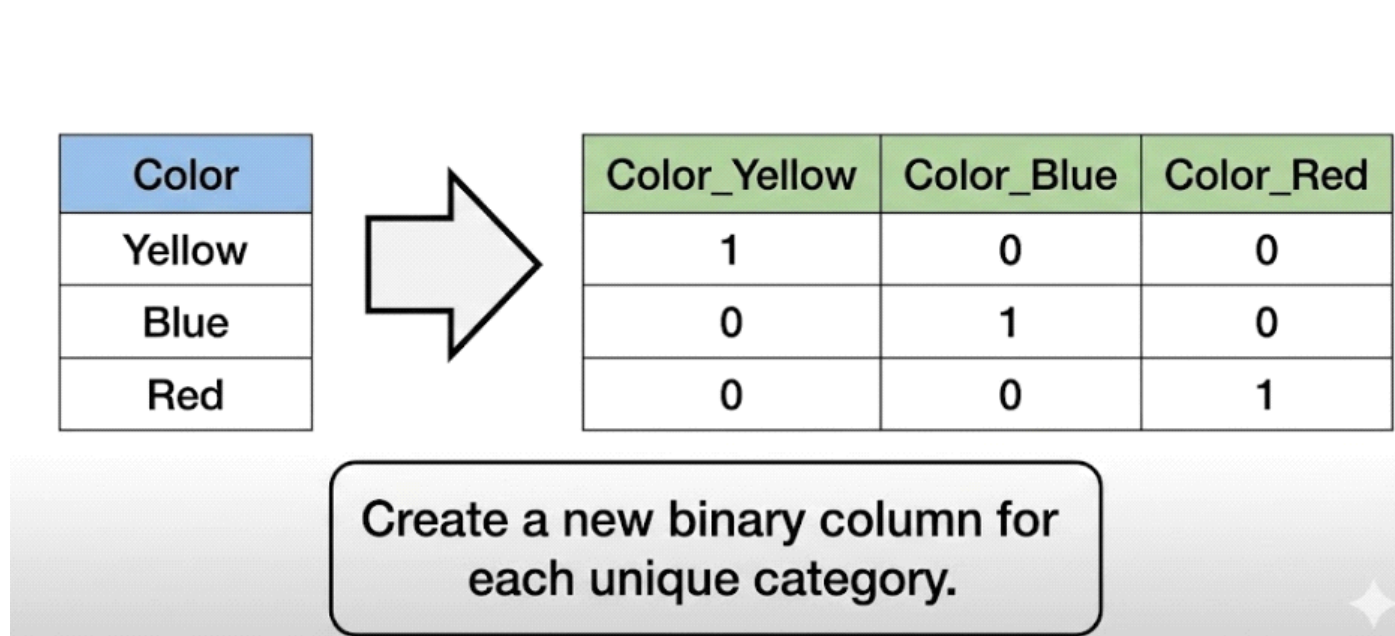
One hot Encoding

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We do not use ordinal in coding because it will assign a number, and let's say red has blue or red has two. A machine will think that red is a more dominant feature.

The Solution: One-Hot Encoding



Understanding the Vectors

Original Value	One-Hot Encoded Vector
Yellow	[1, 0, 0]
Blue	[0, 1, 0]
Red	[0, 0, 1]

This vector means Yellow

This vector means Blue

This vector means Red



Scalability Issue (High Dimensionality)

Category_1	Category_2	...	...	...	...	Category_50
1	0	0	0	0	...	0
0	0	0	1	0	...	0
0	0	0	0	0	...	0
0	0	0	0	1	...	0
0	1	0	0	1	...	0
0	0	0	0	0	...	0
0	0	0	0	0	...	1
0	0	1	0	0	...	0
0	1	0	0	0	...	0
0	0	0	0	0	...	0
0	0	0	0	0	...	0
0	0	0	0	0	...	0
0	0	0	1	0	...	1
0	0	0	0	0	...	0

Yes, you'll get 50 new columns!
This is called the Curse of Dimensionality.

Dummy Variable trap

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The Dummy Variable Trap: Multicollinearity

Color_Yellow	Color_Blue	Color_Red
1	0	0
0	1	0
0	0	1

$$\text{Yellow} + \text{Blue} + \text{Red} = 1$$

PROBLEM:
Perfect
Multicollinearity!
Columns are dependent.

When you use One-Hot Encoding, you create a new column (a "dummy variable") for each category. For example, with three categories (Yellow, Blue, Red), you get three columns. The problem is that these columns are perfectly correlated. As shown below, if you know the values for Blue and Red, you can perfectly predict the value for Yellow (since only one can be 1 at a time). This phenomenon is called **multicollinearity** and is known as the Dummy Variable Trap. It can cause problems for many machine learning algorithms.

The Solution: Drop One Column

Color_Yellow	Color_Blue	Color_Red
Yellow	Blue	Red
Yellow	Blue	Red
Yellow	Blue	Red
Yellow	Blue	Red



Color_Blue	Color_Red
Blue	Red
Blue	Red
Blue	Red
Blue	Red

To avoid the trap, we always drop one column (e.g., the first one).

How N-1 Columns Work (Implicit Representation)

Color_Blue	Color_Red
[1, 0]	[1, 0]
[0, 1]	[0, 1]
[0, 0]	[0, 0]

Means Blue

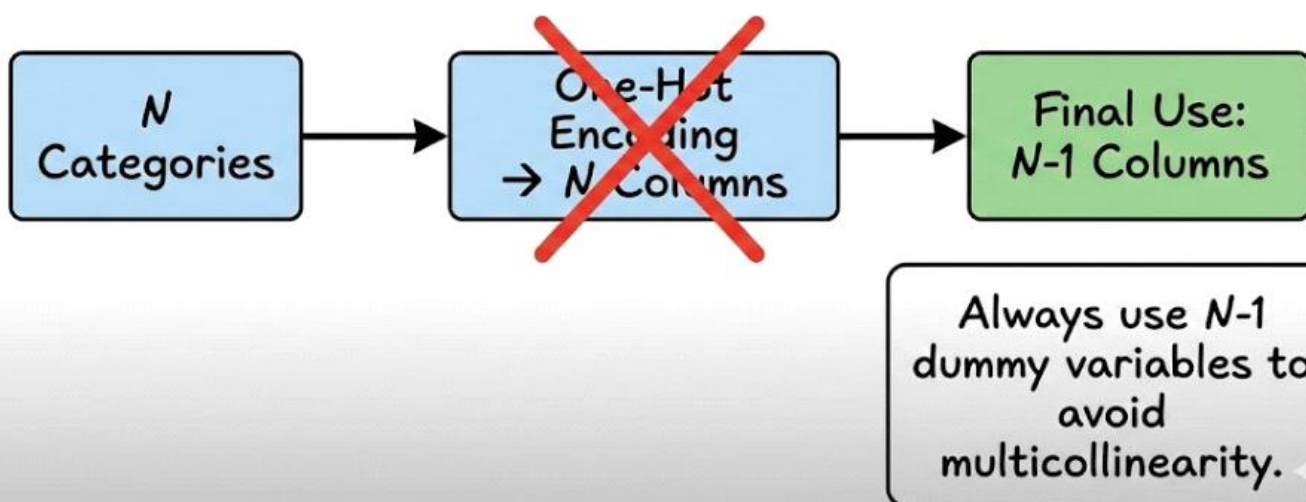
Means Red

Means Yellow (implicitly)

When all remaining columns are 0, it represents the dropped category.

Summary

The General Rule: $N-1$ Dummy Variables



One-Hot Encoding using the most frequent variables

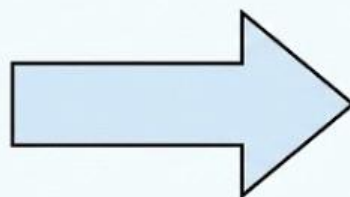
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Car Brand
Maruti Suzuki
Hyundai
Tata
Maruti Suzuki
Hyundai
Land Rover
Jaguar
Audi
Maruti Suzuki
BMW
Mercedes



**Problem: Too Many Columns
(High Dimensionality)**

Original Car Brand
Maruti Suzuki
Hyundai
Tata
Maruti Suzuki
Hyundai
Land Rover
Jaguar
Audi
Maruti Suzuki
BMW
Mercedes



Solution: Group less frequent categories into "Others"

Simplified Car Brand
Maruti Suzuki
Hyundai
Tata
Maruti Suzuki
Hyundai
Others
Others
Others
Maruti Suzuki
Others
Others

	Brand_MarutiSuzuki	Brand_Hyundai	Brand_Tata	Brand_Others
Simplified Car Brand	0	0	0	0
Maruti Suzuki	1	0	0	0
Brand Hyundai	0	1	0	0
Brand Tata	1	0	0	0
Toyota	0	1	0	0
Balil Hyundai	0	0	0	0
Maruti Sunarni	1	0	1	0
Others	0	0	0	1



Final Result: Reduced Dimensionality with One-Hot Encoding